

Design a Notification System - Step 2: Propose High-Level Design and Get Buy-in

Overview

This section discusses the **high-level architecture** of a notification system supporting:

- iOS Push Notifications
- Android Push Notifications
- SMS Messages
- Emails

Main Components Covered

The high-level design is divided into:

1. Different Types of Notifications
2. Contact Information Gathering Flow
3. Notification Sending/Receiving Flow

1. Different Types of Notifications

Overview

Different notification channels use different third-party services and delivery mechanisms.

iOS Push Notification

Main Components

To send iOS push notifications, three major components are required:

1. Provider

Responsibility

- Creates notification request
- Sends request to APNS

Data Required

a. Device Token

- Unique identifier of iOS device
- Used for routing push notification

b. Payload

- JSON dictionary containing notification content

Example Payload

JSON

```
{  
  
  "aps": {
```

```
    "alert": {  
      "title": "New Message",  
      "body": "You received a new message"  
    },  
    "sound": "default"  
  }  
}
```

2. APNS (Apple Push Notification Service)

Description

- Apple-managed remote push notification service

Responsibility

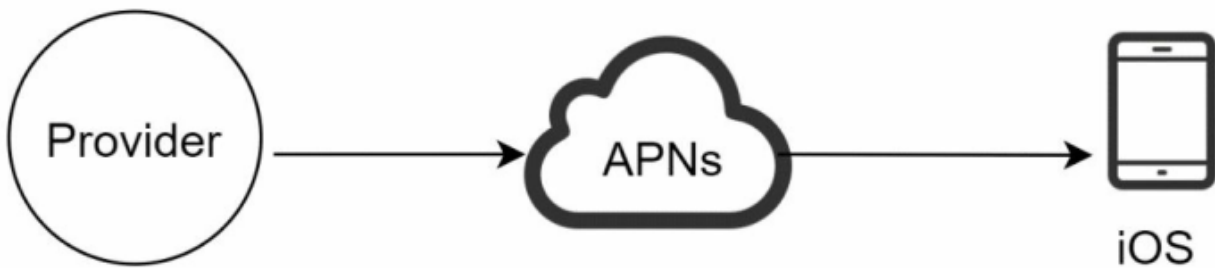
- Receives notifications from provider
- Delivers notifications to iOS devices

3. iOS Device

Description

- End-user device
- Receives and displays push notifications

iOS Push Notification Flow



None

Provider → APNS → iOS Device

Android Push Notification

Overview

Android notification flow is similar to iOS notification flow.

Main Difference

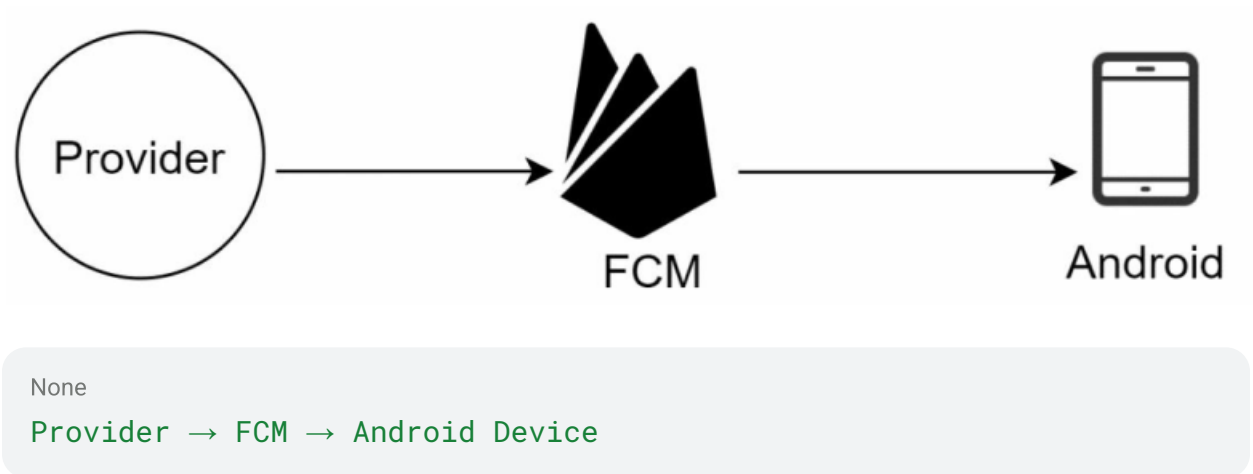
Instead of APNS, Android uses:

Firebase Cloud Messaging (FCM)

Components

- Notification Provider
- FCM Service
- Android Device

Android Push Notification Flow



SMS Message

Overview

SMS notifications are usually handled using third-party SMS providers.

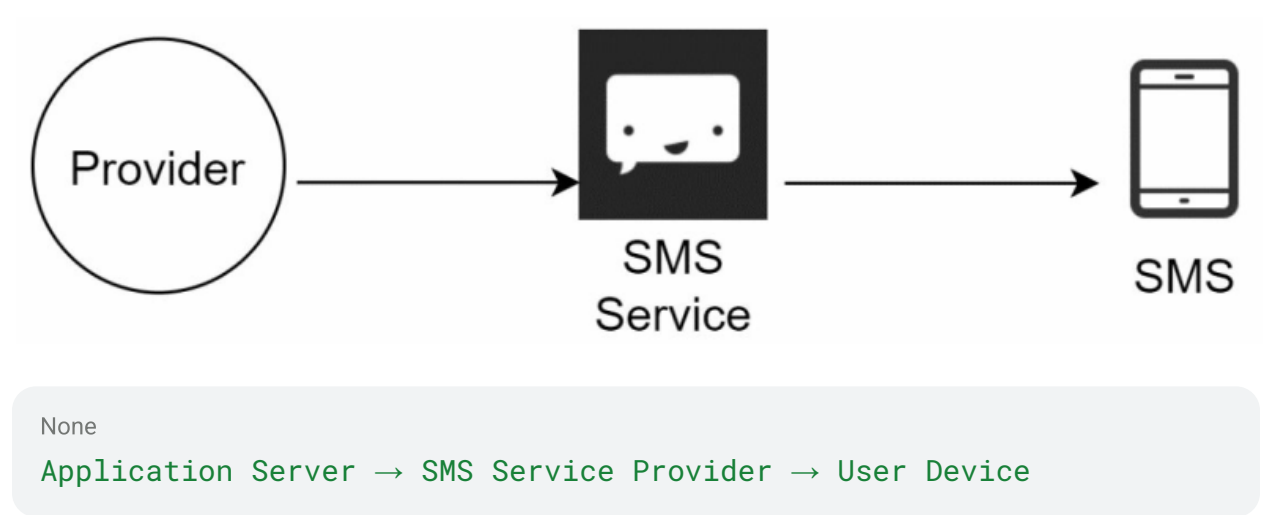
Common SMS Providers

- Twilio
- Nexmo

Characteristics

- Commercial services
- High reliability
- Global delivery support

SMS Notification Flow



Email Notification

Overview

Companies may build email servers themselves, but most prefer third-party email services.

Popular Email Services

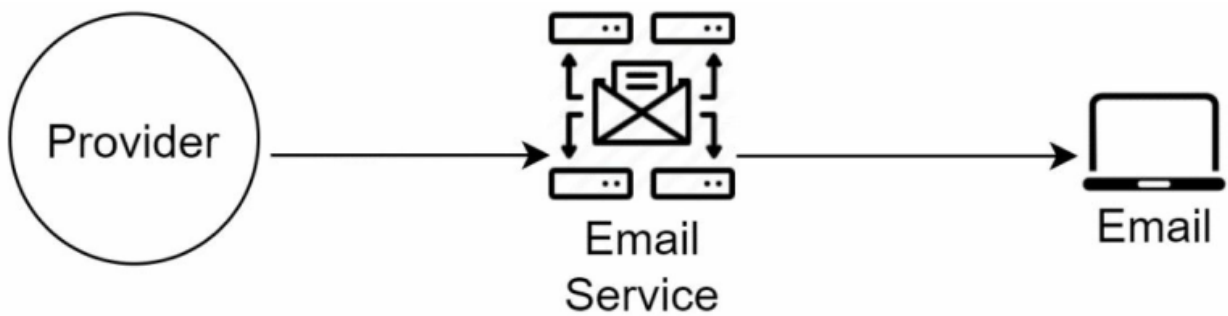
- SendGrid
- Mailchimp

Advantages

- Better email delivery rate
- Analytics support
- Spam filtering

- Scalability

Email Notification Flow



None

Application Server → Email Service Provider → User Mailbox

Combined Notification Architecture

After integrating all notification channels, the system includes:

- Push notification services
- SMS providers
- Email services

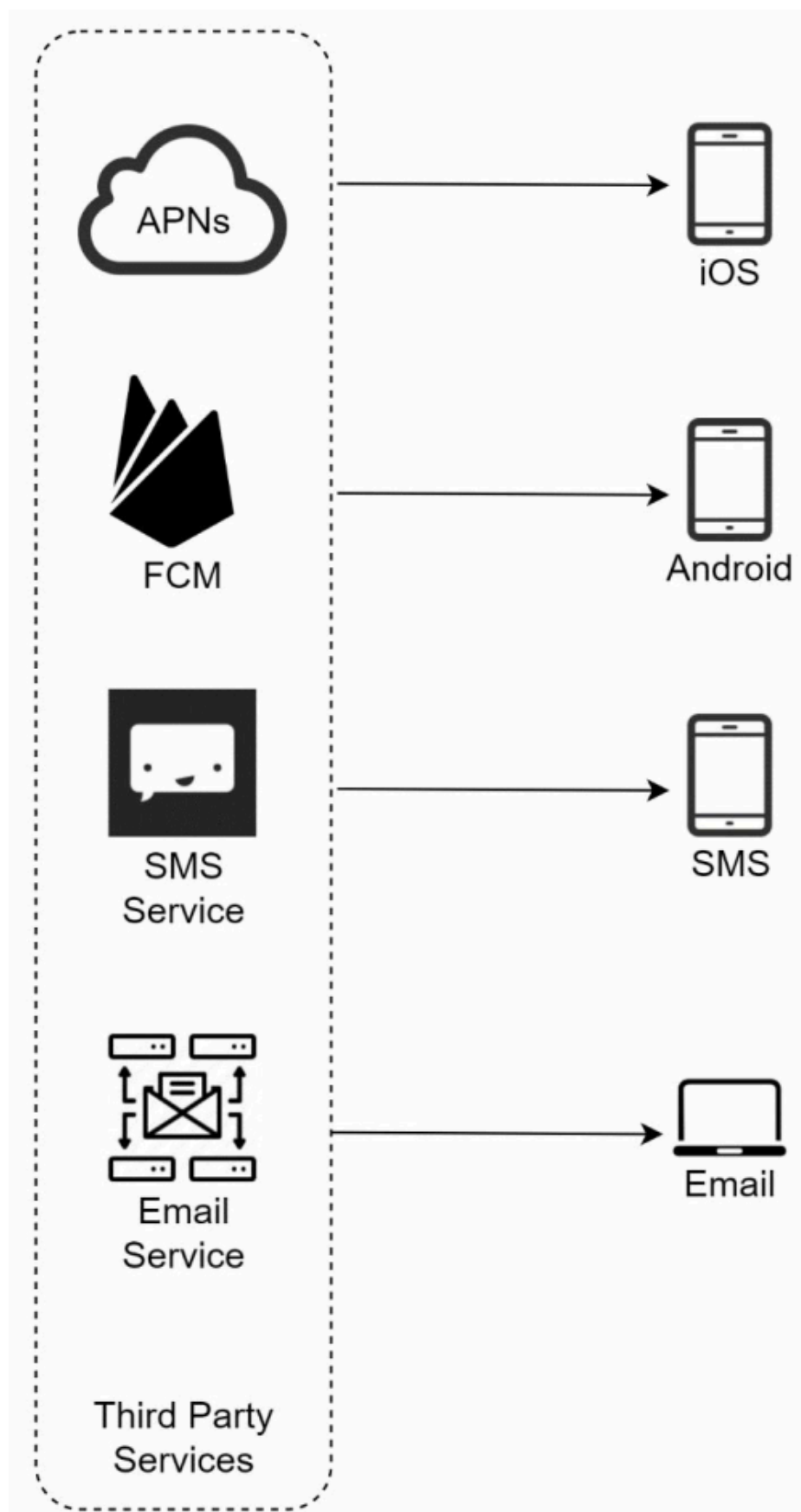


Figure illustrates the complete high-level architecture including all third-party notification services.

Key Design Considerations

1. Multi-channel Support

System supports:

- Push
- SMS
- Email

2. Third-party Integration

External providers handle:

- Message delivery
- Device communication
- Reliability

3. Scalability

Commercial providers help handle:

- Millions of notifications daily

4. Reliability

Notification services provide:

- Retry mechanisms

- Delivery guarantees
- Analytics

Key Takeaway

The notification system architecture relies heavily on:

- External push services
- SMS providers
- Email platforms

Main services used:

Notification Type	Service
iOS Push	APNS
Android Push	FCM
SMS	Twilio / Nexmo
Email	SendGrid / Mailchimp

Goal of High-Level Design:

- Reliable delivery
- Scalability
- Multi-platform support
- Easy integration with third-party services